THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50°	$\angle AOB = 100^\circ$ and OP bisects it
2	**(a)** 30 cm	$PT^2 = OP^2 - OT^2 = 34^2 - 16^2 = 900$; $PT = 30$ cm
3	**(b)** 5 cm	$9 + 7 = AD + 11$; $AD = 5$ cm
4	**(d)** 30°	$\angle OAP = 90^\circ$ (radius $\perp$ tangent); $\angle OAB = 90^\circ - 60^\circ = 30^\circ$
5	**(a)** 41 cm	$OP^2 = r^2 + PT^2 = 9^2 + 40^2 = 1681$; $OP = 41$ cm
6	**(d)** 2 cm	$AF = AB - r = 6 - r$, $CD = BC - r = 8 - r$ and $AF + CD = AC$ (tangent lengths); $(6 - r) + (8 - r) = 10$ gives $r = 2$ cm
7	**(d)** 17 cm	$3x + 5 = 2x + 9$ gives $x = 4$; $QA = 3(4) + 5 = 17$ cm
8	**(c)** $\sqrt{119}$ cm	$PQ^2 = 144 - 25 = 119$
9	**(c)** 28 cm	$PT^2 = OP^2 - OT^2 = 35^2 - 21^2 = 784$; $PT = 28$ cm
10	**(c)** 10 cm	$12 + 8 = AD + 10$; $AD = 10$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tangents from an external point are equal: $QA = QB$; $3x + 4 = 2x + 9$ gives $x = 5$ (1); $QA = 3(5) + 4 = 19$ cm (1).	1+1
12	$PA = PB$, so $\angle PAB = \frac{180^\circ - 40^\circ}{2} = 70^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 70^\circ = 20^\circ$ (1).	1+1
13	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$; $13 + 8 = AD + 9$ (1); $AD = 12$ cm (1).	1+1
14	Tangents from an external point are equal: $QA = QB$; $5x + 1 = 3x + 9$ gives $x = 4$ (1); $QA = 5(4) + 1 = 21$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 12$, $y + z = 7$, $z + x = 9$; adding, $x + y + z = 14$ (1); $AF = 7$ cm, $BD = 5$ cm, $CE = 2$ cm (1).	1+1+1
16	$AC = 29$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 20 - r$, $CD = BC - r = 21 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(20 - r) + (21 - r) = 29$ gives $r = 6$ cm (1).	1+1+1
17	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (1); Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm (1); Chord $= 2 \times 4 = 8$ cm (1).	1+1+1
	OR: Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 14$, $y + z = 10$, $z + x = 12$; adding, $x + y + z = 18$ (1); $AF = 8$ cm, $BD = 6$ cm, $CE = 4$ cm (1).	1+1+1
18	$AC = 40$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 24 - r$, $CD = BC - r = 32 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(24 - r) + (32 - r) = 40$ gives $r = 8$ cm (1).	1+1+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (1); Half-chord $= \sqrt{13^2 - 5^2} = 12$ cm (1); Chord $= 2 \times 12 = 24$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the tangent XY touch the circle with centre O at P, and take any other point Q on XY (1); Q lies outside the circle (a tangent meets the circle only at P), so $OQ > OP$ (1); This holds for every point $Q \neq P$ on XY, so OP is the shortest distance from O to the line XY (2); The shortest distance is the perpendicular, so $OP \perp XY$ (1).	1+1+2+1
	OR: $AC = \sqrt{6^2 + 8^2} = 10$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 6 - r$ and $CD = CE = 8 - r$ and $AE + EC = AC$: $(6 - r) + (8 - r) = 10$ (2); $r = 2$ cm, $AF = 4$ cm, $CD = 6$ cm (1).	2+2+1
20	Let PA and PB be tangents from P to the circle with centre O; join OA, OB, OP (1); $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent) (1); In $\triangle OAP$ and $\triangle OBP$: $OA = OB$ (radii), OP common, right angles equal (1); $\triangle OAP \cong \triangle OBP$ (RHS) (1); Hence $PA = PB$ (CPCT) (1).	1+1+1+1+1
	OR: $AC = \sqrt{5^2 + 12^2} = 13$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 5 - r$ and $CD = CE = 12 - r$ and $AE + EC = AC$: $(5 - r) + (12 - r) = 13$ (2); $r = 2$ cm, $AF = 3$ cm, $CD = 10$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 35°	$\angle OAP = 90^\circ$ (radius $\perp$ tangent); $\angle OAB = 90^\circ - 55^\circ = 35^\circ$
2	**(a)** 45 cm	$OP^2 = r^2 + PT^2 = 27^2 + 36^2 = 2025$; $OP = 45$ cm
3	**(a)** 2 cm	$AF = AB - r = 5 - r$, $CD = BC - r = 12 - r$ and $AF + CD = AC$ (tangent lengths); $(5 - r) + (12 - r) = 13$ gives $r = 2$ cm
4	**(d)** 21 cm	$4x + 1 = 3x + 6$ gives $x = 5$; $QA = 4(5) + 1 = 21$ cm
5	**(d)** 7 cm	$r^2 = 625 - 576 = 49$
6	**(c)** 35 cm	$PT^2 = OP^2 - OT^2 = 37^2 - 12^2 = 1225$; $PT = 35$ cm
7	**(a)** 16 cm	$14 + 11 = AD + 9$; $AD = 16$ cm
8	**(c)** 40°	$\angle OAP = 90^\circ$ (radius $\perp$ tangent); $\angle OAB = 90^\circ - 50^\circ = 40^\circ$
9	**(a)** 50 cm	$OP^2 = r^2 + PT^2 = 14^2 + 48^2 = 2500$; $OP = 50$ cm
10	**(a)** 3 cm	$AF = AB - r = 9 - r$, $CD = BC - r = 12 - r$ and $AF + CD = AC$ (tangent lengths); $(9 - r) + (12 - r) = 15$ gives $r = 3$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$PA = PB$, so $\angle PAB = \frac{180^\circ - 50^\circ}{2} = 65^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 65^\circ = 25^\circ$ (1).	1+1
12	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$; $11 + 12 = AD + 14$ (1); $AD = 9$ cm (1).	1+1
13	Tangents from an external point are equal: $QA = QB$; $4x + 3 = 2x + 11$ gives $x = 4$ (1); $QA = 4(4) + 3 = 19$ cm (1).	1+1
14	$PA = PB$, so $\angle PAB = \frac{180^\circ - 110^\circ}{2} = 35^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 35^\circ = 55^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 12$, $y + z = 11$, $z + x = 13$; adding, $x + y + z = 18$ (1); $AF = 7$ cm, $BD = 5$ cm, $CE = 6$ cm (1).	1+1+1
16	$AC = 39$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 15 - r$, $CD = BC - r = 36 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(15 - r) + (36 - r) = 39$ gives $r = 6$ cm (1).	1+1+1
17	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (1); Half-chord $= \sqrt{10^2 - 6^2} = 8$ cm (1); Chord $= 2 \times 8 = 16$ cm (1).	1+1+1
	OR: Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 10$, $y + z = 9$, $z + x = 15$; adding, $x + y + z = 17$ (1); $AF = 8$ cm, $BD = 2$ cm, $CE = 7$ cm (1).	1+1+1
18	$AC = 35$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 21 - r$, $CD = BC - r = 28 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(21 - r) + (28 - r) = 35$ gives $r = 7$ cm (1).	1+1+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (1); Half-chord $= \sqrt{17^2 - 8^2} = 15$ cm (1); Chord $= 2 \times 15 = 30$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the circle touch AB, BC, CD, DA at P, Q, R, S; tangents from a point are equal: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$ (2); Add: $AB + CD = AD + BC$ (1); In a parallelogram $AB = CD$ and $AD = BC$, so $2AB = 2AD$ (1); $AB = AD$, so all four sides are equal and $ABCD$ is a rhombus (1).	2+1+1+1
	OR: $AC = \sqrt{8^2 + 15^2} = 17$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 8 - r$ and $CD = CE = 15 - r$ and $AE + EC = AC$: $(8 - r) + (15 - r) = 17$ (2); $r = 3$ cm, $AF = 5$ cm, $CD = 12$ cm (1).	2+2+1
20	Let the circle touch AB, BC, CD, DA at P, Q, R, S (1); Tangents from an external point are equal: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$ (2); Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$ (1); So $AB + CD = AD + BC$ (1).	1+2+1+1

	OR: $AC = \sqrt{9^2 + 12^2} = 15$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 9 - r$ and $CD = CE = 12 - r$ and $AE + EC = AC$: $(9 - r) + (12 - r) = 15$ (2); $r = 3$ cm, $AF = 6$ cm, $CD = 9$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90°$; the correct sum is $180°$.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 27 cm	$5x+2=3x+12$ gives $x=5$; $QA=5(5)+2=27$ cm
2	**(a)** 60°	$\angle OQP=90^\circ$
3	**(a)** 36 cm	$PT^2=OP^2-OT^2=39^2-15^2=1296$; $PT=36$ cm
4	**(d)** 5 cm	$8+10=AD+13$; $AD=5$ cm
5	**(c)** 45°	$\angle OAP=90^\circ$ (radius $\perp$ tangent); $\angle OAB=90^\circ-45^\circ=45^\circ$
6	**(d)** 50 cm	$OP^2=r^2+PT^2=30^2+40^2=2500$; $OP=50$ cm
7	**(c)** 3 cm	$AF=AB-r=8-r$, $CD=BC-r=15-r$ and $AF+CD=AC$ (tangent lengths); $(8-r)+(15-r)=17$ gives $r=3$ cm
8	**(a)** 17 cm	$2x+7=x+12$ gives $x=5$; $QA=2(5)+7=17$ cm
9	**(c)** $AB+CD=AD+BC$	Property of a circumscribed quadrilateral
10	**(a)** 32 cm	$PT^2=OP^2-OT^2=40^2-24^2=1024$; $PT=32$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	For a quadrilateral circumscribed about a circle, $AB+CD=AD+BC$; $10+9=AD+6$ (1); $AD=13$ cm (1).	1+1
12	Tangents from an external point are equal: $QA=QB$; $7x+2=4x+14$ gives $x=4$ (1); $QA=7(4)+2=30$ cm (1).	1+1
13	$PA=PB$, so $\angle PAB=\frac{180^\circ-120^\circ}{2}=30^\circ$; $\angle OAP=90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB=90^\circ-30^\circ=60^\circ$ (1).	1+1
14	For a quadrilateral circumscribed about a circle, $AB+CD=AD+BC$; $16+7=AD+12$ (1); $AD=11$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangent lengths from a vertex are equal: $AF=AE=x$, $BD=BF=y$, $CD=CE=z$ (1); $x+y=15$, $y+z=13$, $z+x=14$; adding, $x+y+z=21$ (1); $AF=8$ cm, $BD=7$ cm, $CE=6$ cm (1).	1+1+1
16	$AC=26$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF=AB-r=10-r$, $CD=BC-r=24-r$ and $AF+CD=AC$ (tangent lengths) (1); $(10-r)+(24-r)=26$ gives $r=4$ cm (1).	1+1+1
17	The chord touches the smaller circle at M, so $OM\perp$ chord and M is the midpoint of the chord (1); Half-chord $=\sqrt{25^2-7^2}=24$ cm (1); Chord $=2\times24=48$ cm (1).	1+1+1
	OR: Tangent lengths from a vertex are equal: $AF=AE=x$, $BD=BF=y$, $CD=CE=z$ (1); $x+y=11$, $y+z=8$, $z+x=9$; adding, $x+y+z=14$ (1); $AF=6$ cm, $BD=5$ cm, $CE=3$ cm (1).	1+1+1
18	$AC=45$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF=AB-r=27-r$, $CD=BC-r=36-r$ and $AF+CD=AC$ (tangent lengths) (1); $(27-r)+(36-r)=45$ gives $r=9$ cm (1).	1+1+1
	OR: The chord touches the smaller circle at M, so $OM\perp$ chord and M is the midpoint of the chord (1); Half-chord $=\sqrt{26^2-10^2}=24$ cm (1); Chord $=2\times24=48$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let AB be a diameter and PQ, RS the tangents at A and B (1); $OA\perp PQ$ and $OB\perp RS$ (radius $\perp$ tangent), so $\angle PAB=90^\circ$ and $\angle ABS=90^\circ$ (2); These are alternate interior angles with transversal AB and they are equal (1); Hence $PQ\parallel RS$ (1).	1+2+1+1
	OR: $AC=\sqrt{12^2+16^2}=20$ cm; $BDOF$ is a square of side r, so $BF=BD=r$ (2); $AF=AE=12-r$ and $CD=CE=16-r$ and $AE+EC=AC$: $(12-r)+(16-r)=20$ (2); $r=4$ cm, $AF=8$ cm, $CD=12$ cm (1).	2+2+1
20	Tangents from an external point are equal: $AF=AE$, $BF=BD$, $CD=CE$ (2); Perimeter $=AB+BC+CA=(AF+FB)+(BD+DC)+(CE+EA)$ (1); $=2(AF+BD+CE)$ using the equalities above (1); Hence $AF+BD+CE=\frac{1}{2}(AB+BC+CA)$ (1).	2+1+1+1
	OR: $AC=\sqrt{7^2+24^2}=25$ cm; $BDOF$ is a square of side r, so $BF=BD=r$ (2); $AF=AE=7-r$ and $CD=CE=24-r$ and $AE+EC=AC$: $(7-r)+(24-r)=25$ (2); $r=3$ cm, $AF=4$ cm, $CD=21$ cm (1).	2+2+1

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